

PAPER • OPEN ACCESS

An improved LIO-SAM lidar inertial odometer with dynamic point filtering

To cite this article: Hangtian Zhang *et al* 2024 *J. Phys.: Conf. Ser.* **2902** 012007

View the [article online](#) for updates and enhancements.

You may also like

- [Study on LiDAR inertial tightly coupled SLAM method for outdoor environment](#)
Jiajia Lu, Yue Shen and Xu Wang
- [Visual-LiDAR-Inertial fusion SLAM algorithm for underground detecting robot in coal mines](#)
Yun Bai, Xinyue Liu and Yucheng Li
- [Least squares variance component estimation for robust GNSS/INS/LiDAR integrated with large-scale navigation](#)
Hong Hu, Tanwen Dong, Kerui Liu et al.

An improved LIO-SAM lidar inertial odometer with dynamic point filtering

Hangtian Zhang , Zhongli Ma , Guoliang Zhang and Jiajia Liu *

College of Automation, Chengdu University of Information Technology, Chengdu, Sichuan, 610000, China

* Corresponding author's e-mail: liujj@cuit.edu.cn

Abstract: LIO-SAM is an extensively utilized SLAM (Simultaneous Localization and Mapping) algorithm, featuring an integrated lidar-inertial odometry framework for high-precision, real-time trajectory estimation and mapping in mobile robots. In this paper, we integrate the lidar Iris loop detection algorithm into the LIO-SAM framework to solve the poor loop detection of LIO-SAM itself and the poor loop detection effect of Scan Context resulting from varying lidar directions passing through the same location. Additionally, an enhanced dynamic point filtering algorithm relying on ERASOR is developed to mitigate the presence of dynamic points in the point cloud map created by SLAM. Finally, validation with the KITTI dataset confirms the effectiveness of these algorithms, laying a robust foundation for further advancements.

1. Introduction

Currently, SLAM (Simultaneous Localization and Mapping) [1] is a leading research area of autonomous driving. The sensors related to SLAM mainly include lidar, inertial measurement unit, camera, and other sensors[2], which are basically necessary sensors for autonomous vehicles. The state estimation and mapping method based on IMU-Lidar fusion is widely used.

There are many 3D point cloud mapping algorithms in Lidar SLAM research. The LOAM[3] algorithm which was proposed by Zhang and Singh [3] in 2014 first achieves real-time odometry by splitting SLAM into a high-frequency front-end with lower precision and a low-frequency back-end with higher precision, which meets the requirements of low drift and low complexity. In 2015, the authors proposed a SLAM framework fused with visual sensors called V-LOAM, which is robust in environments lacking feature points and containing fast-moving objects. They added an IMU sensor to V-LOAM in 2018. In 2018, Shan and Englot [4] proposed an improved version of the LeGO-LOAM[4] algorithm based on LOAM, mainly to realize the positioning and mapping of the car in changeable terrain. In this year, Qin et al. [5] proposed a monocular vision-based inertial system VINS-Mono, which implements a tightly coupled vision-based and IMU SLAM algorithm with a camera and a low-cost IMU. LIOM[6], a SLAM algorithm tightly coupled with Lidar-IMU, was proposed by Ye et al. [6], which can output accurate pose estimation values with a high update frequency, optimize the final pose using the rotation constraint algorithm and generate a point cloud map. In 2020, Shan et al. [7] proposed the LIO-SAM algorithm, which adopted the tightly-coupled IMU usage mode on the basis of LOAM, abandoned the inter-frame odometer as the front-end odometer and used the predicted value of the tightly-coupled IMU as the front-end odometer and LEGO-LOAM as the back-end.

This paper studies a LiDAR-inertial odometer using LiDAR and IMU, aiming at some problems existing in LIO-SAM algorithm which is widely used at present, such as the problem of invalid points



and abnormal points in the original point cloud, poor loop detection effect of Scan Context [8] caused by different directions when lidar passes through the same place, and dynamic point in the generated point cloud map after LIO-SAM has built the point cloud map.

2. Related works

2.1. IMU Pre-Integration algorithm

LIO-SAM algorithm needs to calculate the global motion state of the vehicle in the process of mapping. The three coordinate systems and their transformation relationships are used. The coordinate system transformation diagram is depicted in Figure 1.

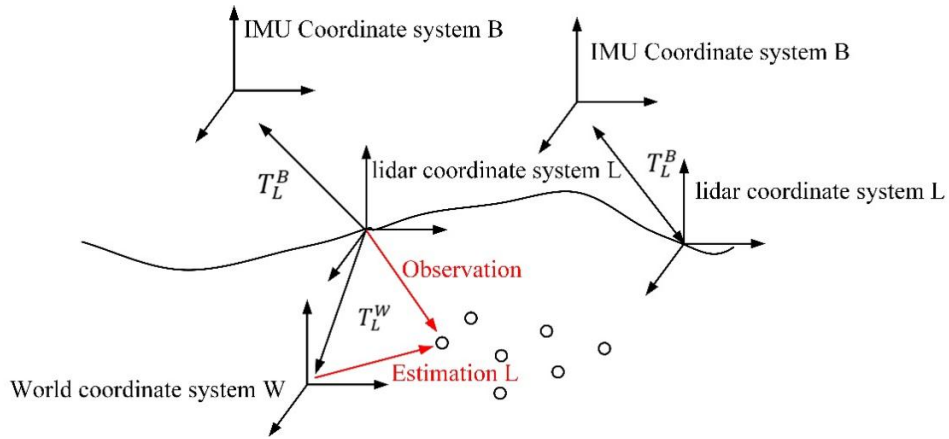


Figure 1. Coordinate system transformation diagram.

The IMU measures angular velocity and acceleration at the current time t , as defined by Equations (1) and (2) [10]:

$$\hat{w}_t = w_t + b_t^w + n_t^w \quad (1)$$

$$\hat{a}_t = R_t^{BW}(a_t - g) + b_t^a + n_t^a \quad (2)$$

where $\hat{w}_t$ is the gyroscope value obtained in the B coordinate frame; w_t is the ideal angular velocity obtained in the B coordinate frame; b_t^w is the zero bias value of the gyroscope in the B coordinate frame at time t ; n_t^w denotes the white noise from the gyroscope in the B coordinate frame at time t . Similarly, the representation of $\hat{a}_t$ is the same; R_t^{BW} is the rotation matrix at time t transformed from W coordinate frame to B coordinate frame; g represents gravitational acceleration obtained from the W frame.

Integrating the IMU's angular velocity, acceleration, and rotation over the interval $[t, t + \Delta t]$ can obtain the displacement $p_{t+\Delta t}$, velocity $v_{t+\Delta t}$ and rotation matrix $R_{t+\Delta t}$ at time $t + \Delta t$ as follows:

$$v_{t+\Delta t} = v_t + g\Delta t + R_t(\hat{a}_t - b_t^a - n_t^a)\Delta t \quad (3)$$

$$p_{t+\Delta t} = p_t + v_t\Delta t + \frac{1}{2}g\Delta t^2 + \frac{1}{2}R_t(\hat{a}_t - b_t^a - n_t^a)\Delta t^2 \quad (4)$$

$$R_{t+\Delta t} = R_t \exp((\hat{w}_t - b_t^w - n_t^w)\Delta t) \quad (5)$$

where $R_t = R_t^{WB} = R_t^{BW^T}$. We apply the IMU integration method from [10] to determine the relative motion between two time steps. The pre-integrated measurements are Δv_{ij} , Δp_{ij} and ΔR_{ij} between time i and j .

2.2. Lidar iris loop detection

The name Lidar Iris originates from human Iris signature for identification, which generates a binary feature map from Lidar Iris [11] from every frame of point cloud. Therefore, the algorithm creates a

historical database of Lidar Iris binary feature maps as the unmanned vehicle moves through the scene. The generation of Lidar Iris descriptors can be theoretically decomposed into three steps: First, Lidar Iris images are generated. Second, Fourier transforms the Lidar Iris image. The third is to perform binarization on the image.

In this paper, we integrate the Lidar Iris loop detection algorithm into the LIO-SAM framework, addressing the large errors of the Euclidean distance-based loop detection used in LIO-SAM.

2.3. ERASOR dynamic point filtering

After the point cloud map is built by SLAM, it is inevitable to encounter the interference of dynamic objects, which can form "ghosting" within the point cloud map. The dynamic points can be removed by post-processing the map once with the ERASOR algorithm. The framework diagram of ERASOR refers to Figure 2 [9].

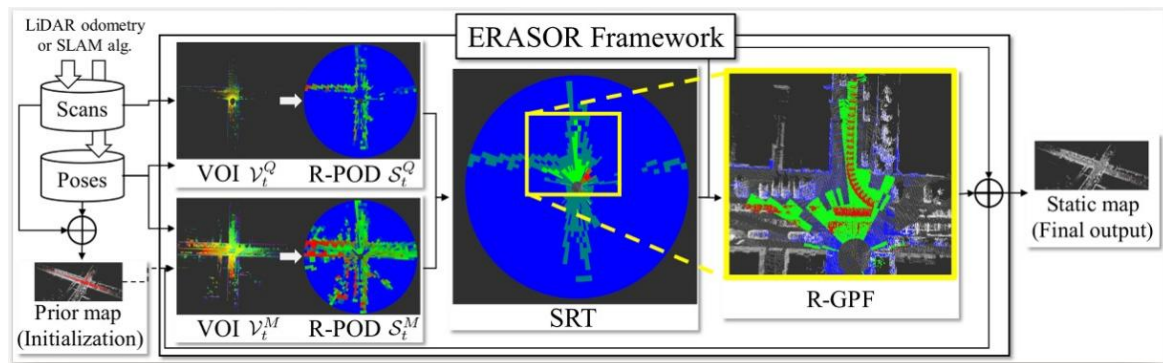


Figure 2. ERASOR Frame diagram [9].

Algorithmic flow for ERASOR can be summarized as four parts: (1) We circle the submap corresponding to query scan and divide them into bins in the same way. (2) We compute the occupancy state descriptor (point cloud height difference) for each bin. (3) We find those bins in the submap whose descriptor ratio is less than 0.2 as potential dynamic bins. (4) Ground plane is fitted in each potential dynamic bin, and above-ground points are filtered as moving points. In this paper, the ERASOR algorithm is improved and integrated into the last step of dynamic point filtering of the final generated map.

3. Improved LIO-SAM Lidar inertial odometry

3.1. System overview

Designing a system based on LIO-SAM is displayed in Figure 3. First, the original point cloud data was necessarily filtered to filter out the noise points. Then, the Lidar Iris loop detection algorithm is included in graph optimization to adjust deviations, which solves the trajectory drift problem caused by the accumulated motion error of the loop detection algorithm using Euclidean distance in LIO-SAM.

3.2. Improved dynamic point filtering based on ERASOR

In this paper, after the point cloud map is formed by LIO-SAM, the map is processed once by the ERASOR algorithm to discard dynamic points from the map. The ERASOR algorithm only considers the space between -1m and 3m as the Volume of Interest (VOI). The algorithm first needs to convert a local submap P_t^M circled by the original point cloud map containing the dynamic object and the original point cloud P_t^Q at time t into an aerial view respectively. The point cloud frame's center is chosen as the circle's center, which is then divided into N_s equidistant segments according to the aerial view angle. The bird's-eye view radius is L_{max} . The number of rings is set to N_r .

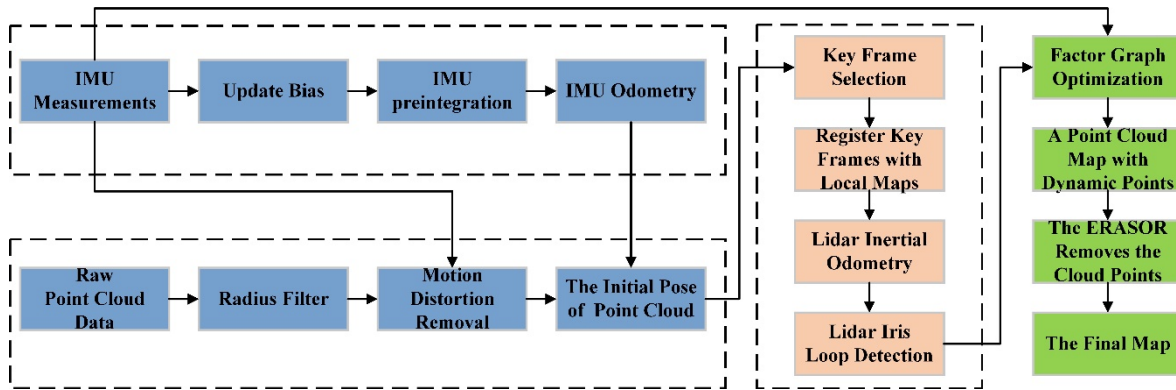


Figure 3. System structure.

Then the distance between the rings is $\frac{L_{max}}{N_r}$, the angle of a single sector area is $\frac{360^\circ}{N_s}$, and N_r in ERASOR algorithm is 20. The length of the small sector obtained by this partition method is 1/20 of the radius of the overhead view. The ring spacing is set to the radius difference of concentric circles formed by adjacent laser beams and the ground plane $\Delta r_{i,i+1}$, which is calculated as shown in Equation (6):

$$\Delta r_{i,i+1} = \frac{\tan \alpha_{i+1} - \tan \alpha_i}{\tan \alpha_i \tan \alpha_{i+1}} h \quad (6)$$

where h is the Lidar height above the ground, the angle of i th laser emitter and the ground plane is α_i , and the angle between the i th plus laser emitter and the ground plane is α_{i+1} .

The spacing of the concentric circles formed by the ground plane is $\Delta r_{i,i+1}$ as the spacing between the rings, and the spacing is l_{n-2} ($n=3\sim 22$), respectively, as shown in Equation (7), to guarantee a relatively uniform distribution of point clouds in each sector.

$$l_{n-2} = \begin{cases} \Delta r_{3n-2,3n+1} = \frac{\tan \alpha_{i+1} - \tan \alpha_i}{\tan \alpha_i \tan \alpha_{i+1}} h, n = 3 \sim 21 \\ r_{lines} = \frac{h}{\tan \alpha_{lines}}, n = 22 \end{cases} \quad (7)$$

The average of the three points with the largest Z coordinate in each bin is used as the maximum value. Then the average of the three points with the smallest Z coordinate in each bin is used as the minimum value. The variation between the above calculated highest and lowest values is taken as the eigenvalue of the bin. If there are fewer than six points in the bin, we calculate the maximum z-coordinate of all the points and the minimum z-coordinate of all the points. We take the variation between the maximum value and the minimum value as the feature value of the bin.

4. Experimental results and analysis

4.1. Experimental platform and parameter setting

The experimental system platform is ubuntu18.04LTS, the kernel version is Linux version 5.4.0-89-generic, the point cloud processing library is PCL1.8, and the back-end factor optimization library is Gtsam4.0.2. The hardware configuration environment is GeForce GTX 1050 Ti Intel(R) Core(TM) i5-7300HQ CPU @ 2.50GHz. The KITTI dataset is used in this paper.

4.2. Analysis of lidar iris loop detection algorithm

Regarding the definition of correct loopback, we consider that the distance between the query node and the matching node in the trajectory truth value is less than 4m as the correct loopback. Using this metric, we calculate 332 correct loops for the 2011_09_30_drive_0028 dataset, as shown in Figure 4.

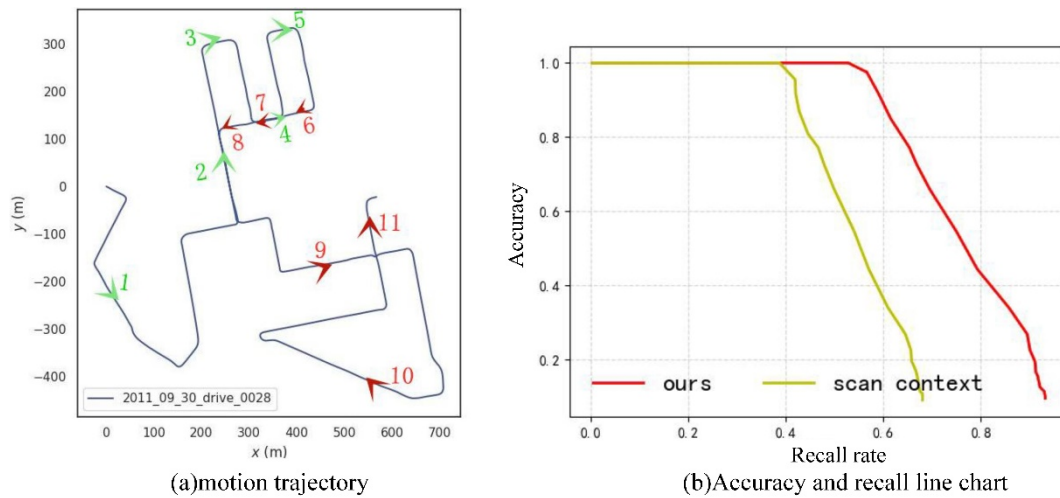


Figure 4. Movement track chart (a) and line chart of precision and recall rate (b).

Figure 4(a) shows the motion trajectory of KITTI 2011_09_30_drive_0028 dataset, where the green arrow represents the motion trajectory without repetition at the beginning, and the red arrow represents the repeated motion trajectory at the last segment. The arrow direction represents the motion direction and the number represents the motion sequence. The 2011_09_30_drive_0028 dataset contains reverse closed loops and closed loops with different directions at the intersection.

Figure 4(b) shows the line chart of precision and recall rate of the Scan Context algorithm and the proposed algorithm. The maximum recall rate of Scan Context can reach 0.3886 when the accuracy rate is 1, and the maximum recall rate of the proposed algorithm can reach 0.5301 when the accuracy rate is 1. As the distance threshold in loop detection is increased, the accuracy rate decreases, but the recall rate of the proposed algorithm consistently exceeds that of Scan Context.

4.3. Evaluation of improved ERASOR dynamic point filtering

Figure 5(a) illustrates the scene diagram of a road segment from the KITTI dataset 2011_09_30_drive_0027. Figure 5(b) presents the initial point cloud map containing dynamic points. The green points depicted in Figure 5(c) represent dynamic points filtered out by our proposed algorithm. Finally, Figure 5(d) displays the static map generated by our method after removing these dynamic elements, demonstrating that our algorithm effectively eliminates both vehicular motion trajectories and individual moving entities.

We also tested it on KITTI 2011_09_26_drive_0014. This data represents vehicles driving on the highway. The final map produced by LIO-SAM has the "ghost" and its own trajectory. The dynamic points are successfully removed by the proposed algorithm (Figure 6).

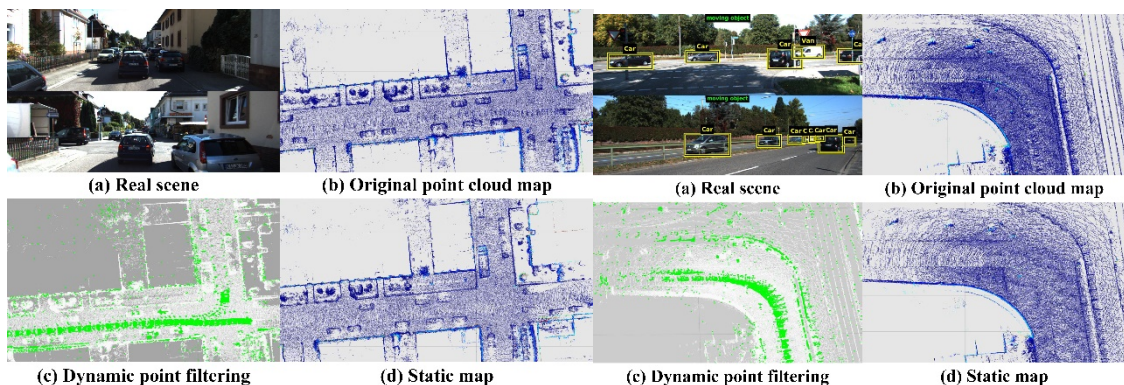


Figure 5. Dynamic point filtering effect 1. **Figure 6.** Dynamic point filtering effect 2.

4.4. Comparative experiment

The data set given by LIO-SAM is selected for testing. Firstly, the proposed algorithm is compared with FAST-LIO2 [12]. Figure 7 shows the comparison under CloudCompare software.

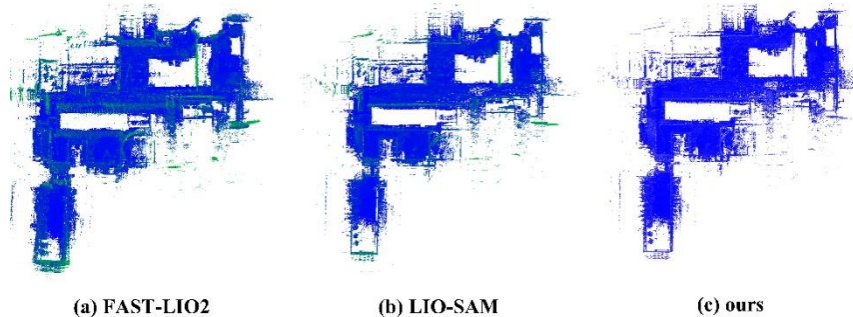


Figure 7. Overall comparison.

It can be seen that the first two maps have inaccurate point cloud registration with multiple shadows. Our method performs better.

A detailed comparison with LIO-SAM is illustrated in Figure 8. Our algorithm demonstrates superior positioning and mapping capabilities during motion.

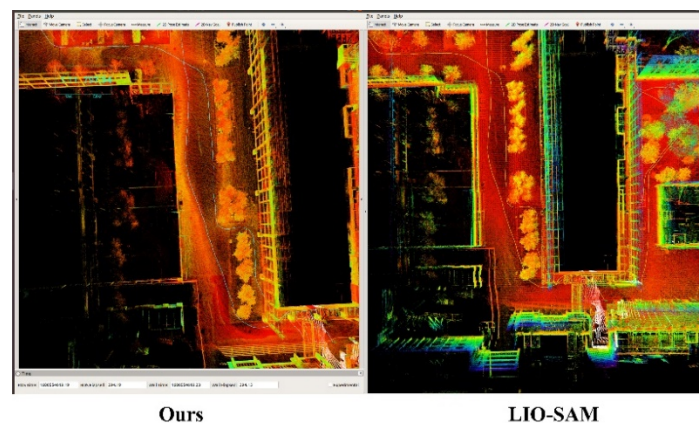


Figure 8. Detail comparison.

5. Conclusion

This paper proposes an enhancement to the LIO-SAM algorithm by applying radius-based filtering to the original point cloud data, thereby reducing noise within the dataset. To address the issue of suboptimal loop detection in Scan Context when Lidar revisits locations from different orientations, we integrate the Lidar Iris loop detection algorithm into the LIO-SAM framework. Experimental comparisons demonstrate that our proposed algorithm achieves improved precision and recall rates relative to its predecessor. Additionally, enhancements have been made to the ERASOR algorithm. By evaluating several common dynamic point filtering algorithms across various scenarios, we validate that our method effectively filters out dynamic points from maps containing them and generates a final 3D high-precision map devoid of such points.

Acknowledgment

The authors are grateful to the Key R&D project of Science and Technology Department of Sichuan Province, under Grant (2023YFG0196 and 2023YFN0077), Science and Technology achievements transformation Project of Science and Technology Department of Sichuan Province, under Grant (2023JDZH0023), Sichuan Provincial Science and Technology Department, Youth Fund project, under Grant (2023NSFSC1429).

References

- [1] Tian C, Liu H, Liu Z, et al. Research on multi-sensor fusion SLAM algorithm based on improved gmapping [J]. *IEEE Access*, 2023, 11: 13690-13703.
- [2] Wang Zihao. Research on high-precision map generation algorithm based on GNSS/IMU/LiDAR [D], 2021.
- [3] Zhang J, Singh S. LOAM: Lidar Odometry and Mapping in Real-time [C]//*Robotics: Science and Systems*. 2014, 2(9): 1-9.
- [4] Shan T, Englot B. LeGO-LOAM: Lightweight and Ground-Optimized Lidar Odometry and Mapping on Variable Terrain [C]. *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2018: 4758-4765.
- [5] Qin T, Li P, Shen S. Vins-mono: A robust and versatile monocular visual-inertial state estimator [J]. *IEEE Transactions on Robotics*, 2018, 34(4): 1004-1020.
- [6] Ye H, Chen Y, Liu M. Tightly Coupled 3D Lidar Inertial Odometry and Mapping [J]. *IEEE*, 2019:3144-3150.
- [7] Shan T, Englot B, Meyers D, et al. Lio-sam: Tightly-coupled lidar inertial odometry via smoothing and mapping [C]//2020 IEEE/RSJ international conference on intelligent robots and systems (IROS). *IEEE*, 2020: 5135-5142.
- [8] Kim G, Kim A. Scan context: Egocentric spatial descriptor for place recognition within 3d point cloud map [C]//2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). *IEEE*, 2018: 4802-4809.
- [9] Lim H, Hwang S, Myung H. ERASOR: Egocentric ratio of pseudo occupancy-based dynamic object removal for static 3D point cloud map building [J]. *IEEE Robotics and Automation Letters*, 2021, 6(2): 2272-2279.
- [10] Zhou Q, Li Z, Yu G, et al. A novel adaptive Kalman filter for Euler-angle-based MEMS IMU/magnetometer attitude estimation [J]. *Measurement Science and Technology*, 2021, 32(4): 045104.
- [11] Wang Y, Sun Z, Xu C Z, et al. Lidar iris for loop-closure detection [C]//2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). *IEEE*, 2020: 5769-5775.
- [12] Xu W, Cai Y, He D, et al. Fast-lid2: Fast direct lidar-inertial odometry [J]. *IEEE Transactions on Robotics*, 2022, 38(4): 2053-2073.